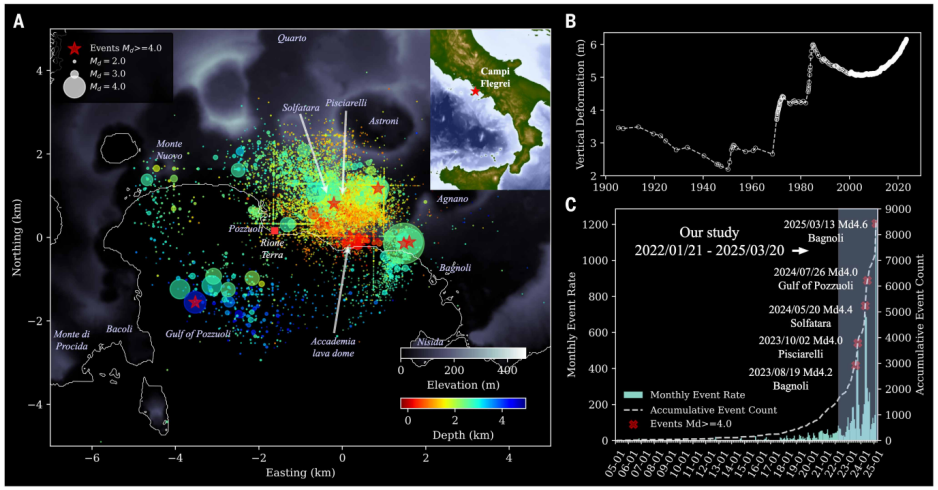
# Figure 1: Seismicity and ground displacement

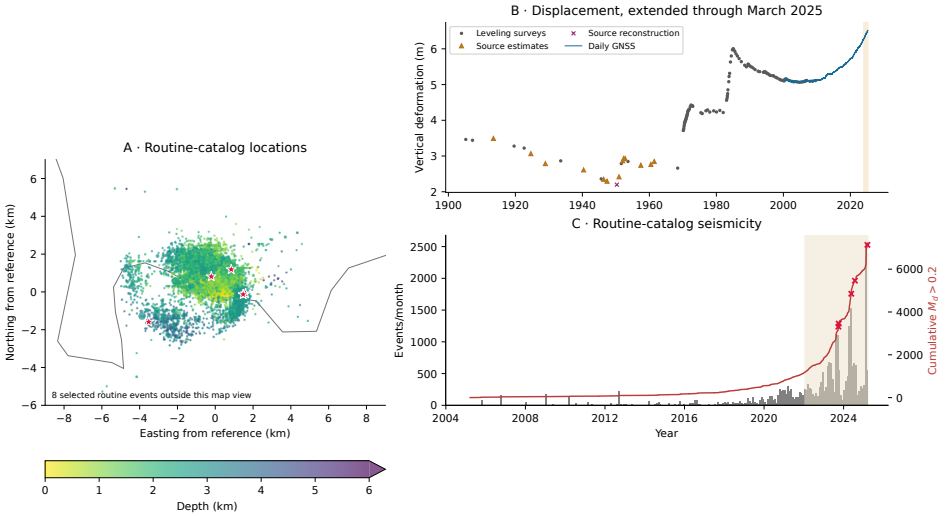
## Published reference




Tan et al. (2025), tanmodel.pdf, p. 2.
DOI: 10.1126/science.adw9038.

## Independent result

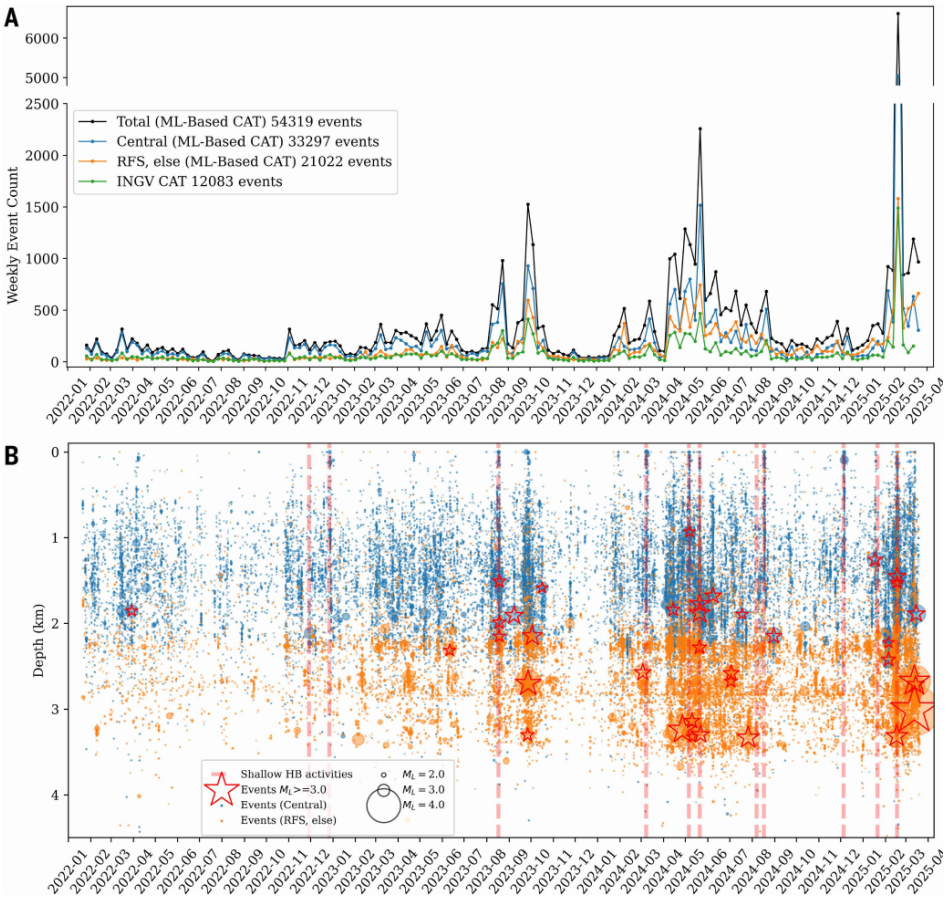
### Panel 1A–C — partially reproduced




Partial contextual reconstruction. A: external routine catalog, 16,046 located events and five Md>=4 stars. B: 114 leveling surveys (circles), 13 source-provided estimates (triangles), one source-provided reconstruction (cross), and daily GNSS through 20 March 2025; extension beyond 2023 shaded. Historical estimates are not raw observations or reconstructions made here. C: all routine monthly counts and cumulative Md>0.2; March includes days 1–20 only. Routine events are not independent detections. Datum joining is an assumption; basemap and live snapshot differ from the publication.

# Figure 2: Temporal evolution of seismicity

## Published reference




Tan et al. (2025), tanmodel.pdf, p. 3.

DOI: 10.1126/science.adw9038.

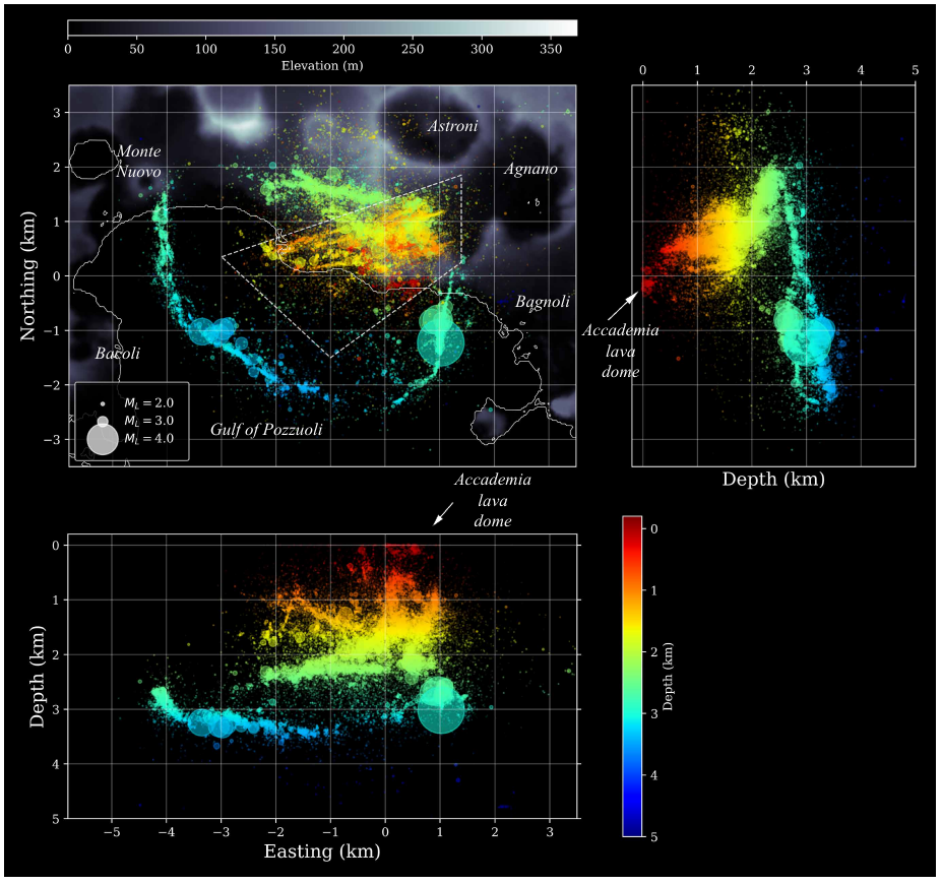
## Independent result

**All panels/views — not reproduced**

Panels A and B need multiyear calibrated/relocated events and magnitudes. The current 908 initial associations remain provisional.

# Figure 3: Spatial distribution of seismicity

## Published reference




Tan et al. (2025), tanmodel.pdf, p. 4.
DOI: 10.1126/science.adw9038.

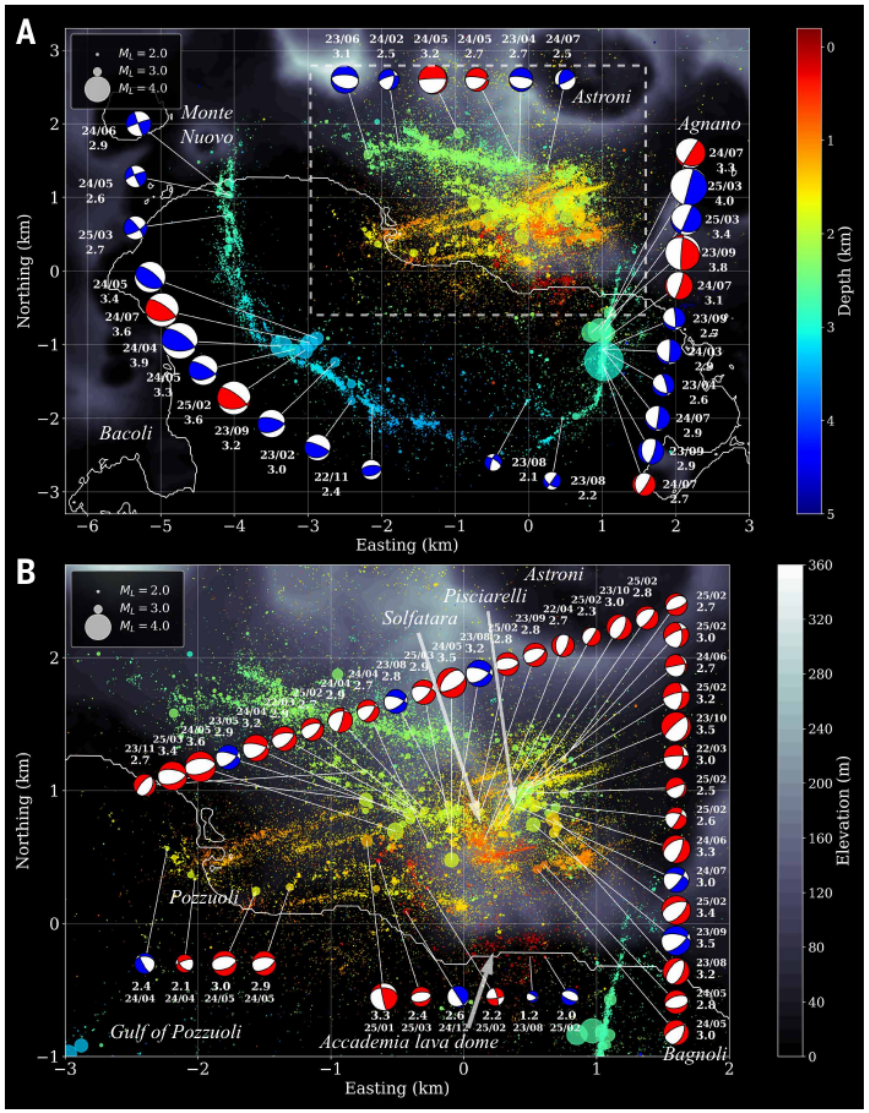
## Independent result

**All panels/views — not reproduced**

The map and both cross sections require independently calibrated and double-difference-relocated events. Provisional diagnostic locations are not substitutes.

# Figure 4: Focal mechanisms of larger events

## Published reference




Tan et al. (2025), tanmodel.pdf, p. 5.
DOI: 10.1126/science.adw9038.

## Independent result

**All panels/views — not reproduced**

Panels A and B require independent SKHASH/gCAP solutions and reliable locations. No focal inversion was executed.

# Figure S1: Recorded station availability

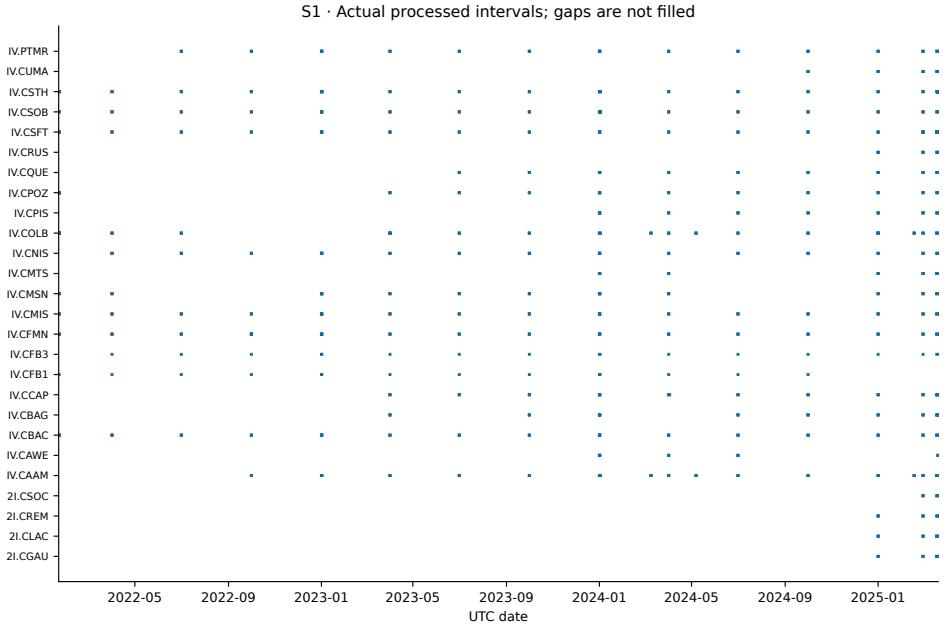
## Published reference




Tan et al. (2025), tanmodelsupp.pdf, p. 12.
DOI: 10.1126/science.adw9038.

## Independent result

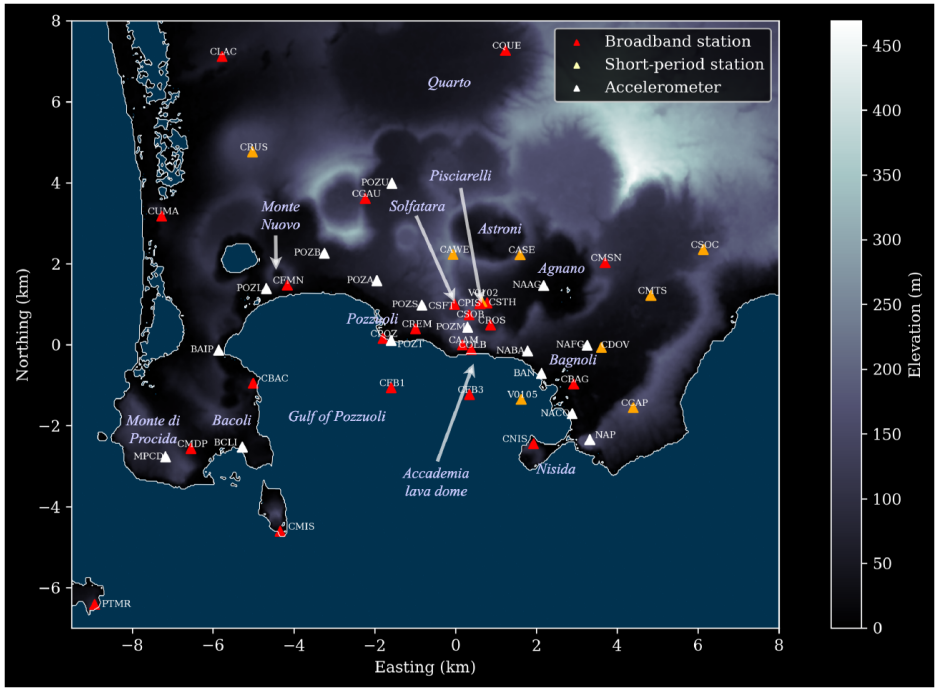
### Panel S1 — partially reproduced




Partial: 608 processed files, 26 station names and 4278.694 channel-hours after union of overlapping intervals. Empty intervals are not connected or counted as coverage.

# Figure S2: Station locations

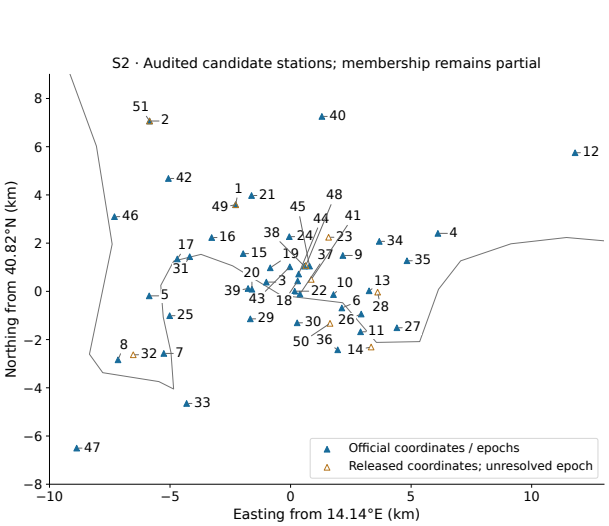
## Published reference




Tan et al. (2025), tanmodelsupp.pdf, p. 13.
DOI: 10.1126/science.adw9038.

## Independent result

### Panel S2 — partially reproduced



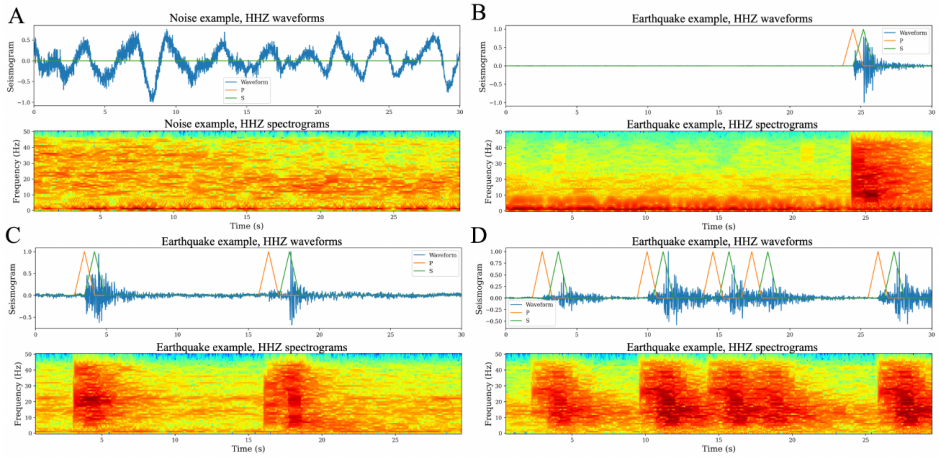

| | | | |
|---|---|---|---|
| 1 | 2I.CGAU..HH | 27 | IV.CCAP..EH |
| 2 | 2I.CLAC..HH | 28 | IV.CDOV..EH |
| 3 | 2I.CREM..HH | 29 | IV.CFB1..HH |
| 4 | 2I.CSOC..EH | 30 | IV.CFB3..HH |
| 5 | IT.BAIP..HN | 31 | IV.CFMN..HH |
| 6 | IT.BAN..HN | 32 | IV.CMDP..HH |
| 7 | IT.BCLI..HN | 33 | IV.CMIS..HH |
| 8 | IT.MPCD..HN | 34 | IV.CMSN..HH |
| 9 | IT.NAAG..HN | 35 | IV.CMTS..EH |
| 10 | IT.NABA..HN | 36 | IV.CNIS..HH |
| 11 | IT.NACO..HN | 37 | IV.COLB..HH |
| 12 | IT.NAD..HN | 38 | IV.CPIS..HH |
| 13 | IT.NAFG..HN | 39 | IV.CPOZ..HH |
| 14 | IT.NAP..HN | 40 | IV.CQUE..HH |
| 15 | IT.POZA..HN | 41 | IV.CROS..HH |
| 16 | IT.POZB..HN | 42 | IV.CRUS..EH |
| 17 | IT.POZL..HN | 43 | IV.CSFT..HH |
| 18 | IT.POZM..HN | 44 | IV.CSOB..HH |
| 19 | IT.POZS..HN | 45 | IV.CSTH..HH |
| 20 | IT.POZT..HN | 46 | IV.CUMA..HH |
| 21 | IT.POZU..HN | 47 | IV.PTMR..HH |
| 22 | IV.CAAM..HH | 48 | IV.V0102..EH |
| 23 | IV.CASE..EH | 49 | IV.V0104..HH |
| 24 | IV.CAWE..EH | 50 | IV.V0105..EH |
| 25 | IV.CBAC..HH | 51 | IV.V0106..HH |
| 26 | IV.CBAG..HH | | |

42 official epoch matches; 9 unresolved.
51 identifiers ≠ verified 48 instruments.
Two coordinate-coincident pairs are retained.
See station_geometry_qa.csv for every entry.

Partial station inventory: 51 released identifiers audited against official study-period response epochs. Filled triangles: 42 official coordinate matches; open triangles: 9 released coordinates with unresolved epochs or channel codes. Two coincident-coordinate pairs remain distinct identifiers. No verified geographic outlier. Local WGS84 projection and clipped coastline; the original 48-instrument membership is not established.

# Figure S3: Training examples

## Published reference




Tan et al. (2025), tanmodelsupp.pdf, p. 14.
DOI: 10.1126/science.adw9038.

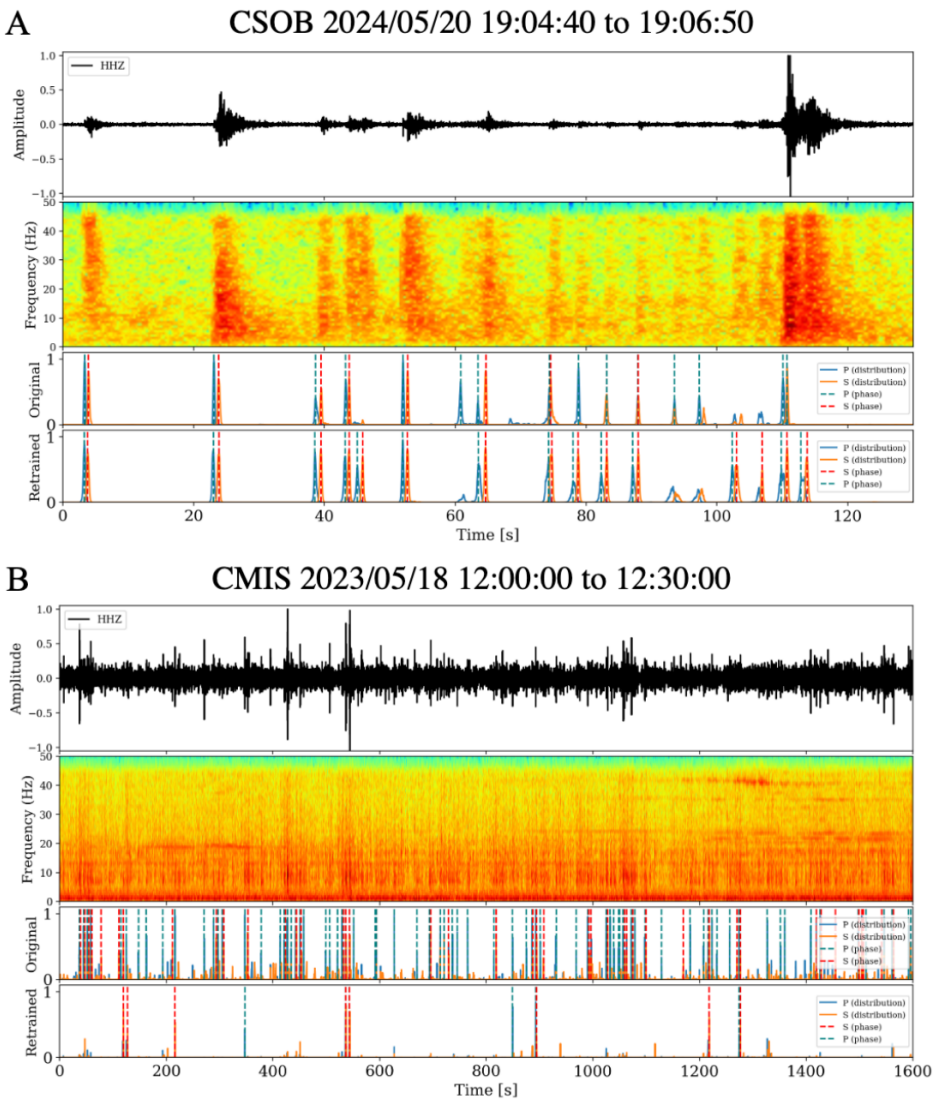
## Independent result

**All panels/views — not reproduced**

Panels A–D require the specified training traces, annotations and augmentation examples. These were not reconstructed; downstream use of trained weights is not retraining.

# Figure S4: PhaseNet benchmark examples

## Published reference




Tan et al. (2025), tanmodelsupp.pdf, p. 15.
DOI: 10.1126/science.adw9038.

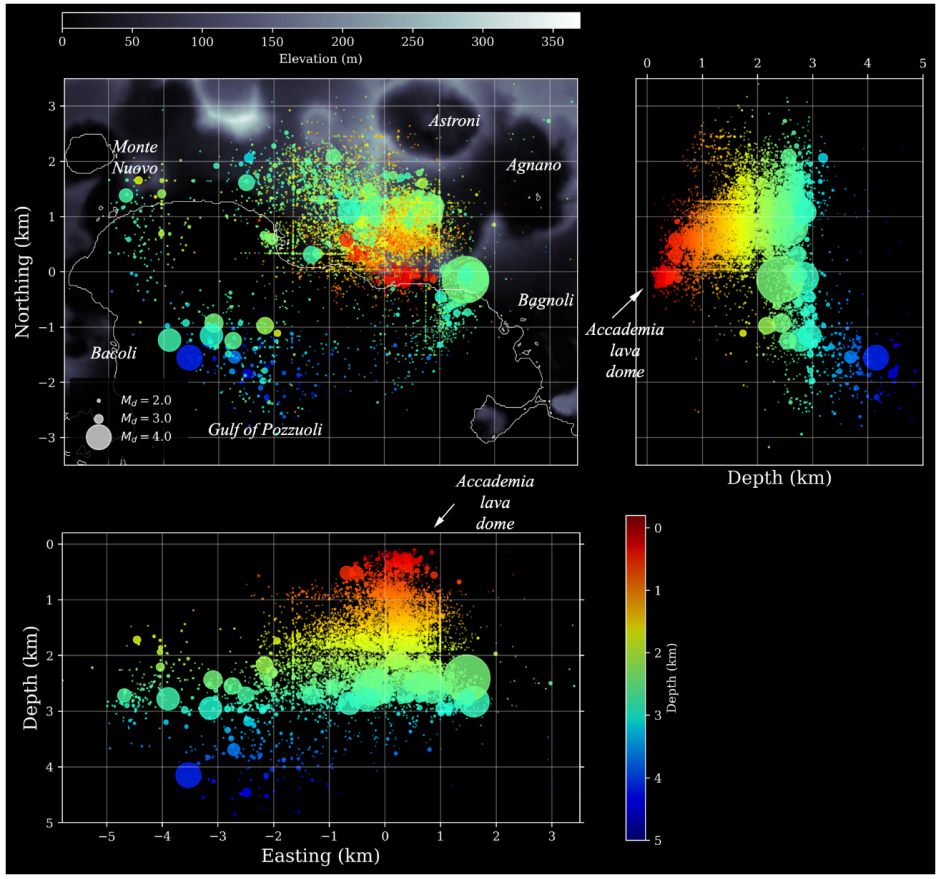
## Independent result

**All panels/views — not reproduced**

Panels A and B were not generated for the exact CSOB swarm and CMIS noise examples. Other unlabeled inference comparisons remain diagnostics.

# Figure S5: Routine-catalog locations

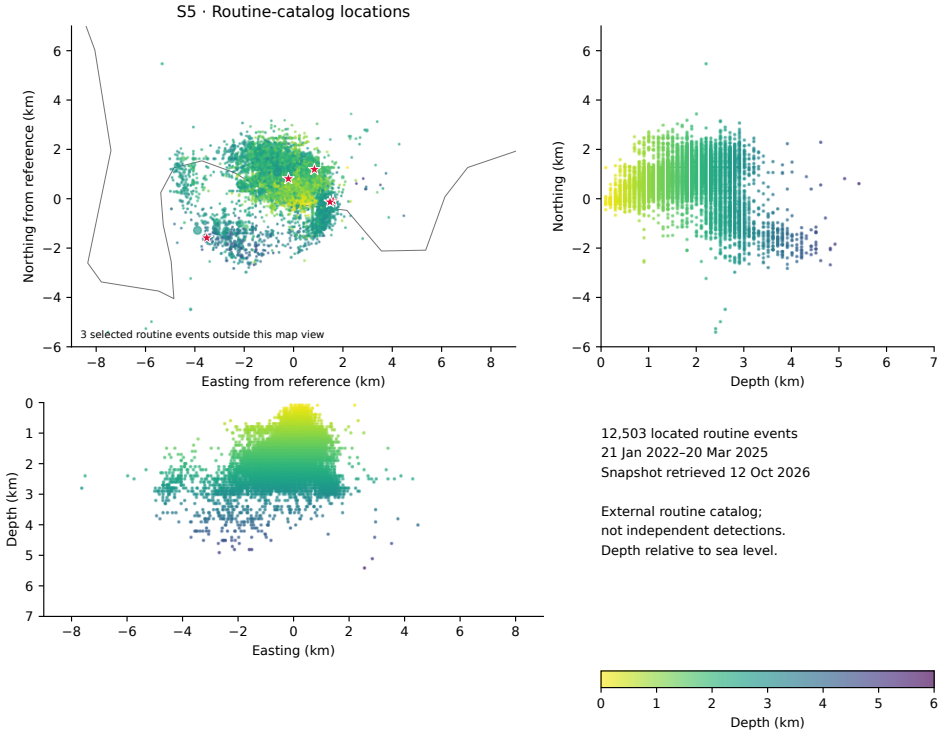
## Published reference




Tan et al. (2025), tanmodelsupp.pdf, p. 16.
DOI: 10.1126/science.adw9038.

## Independent result

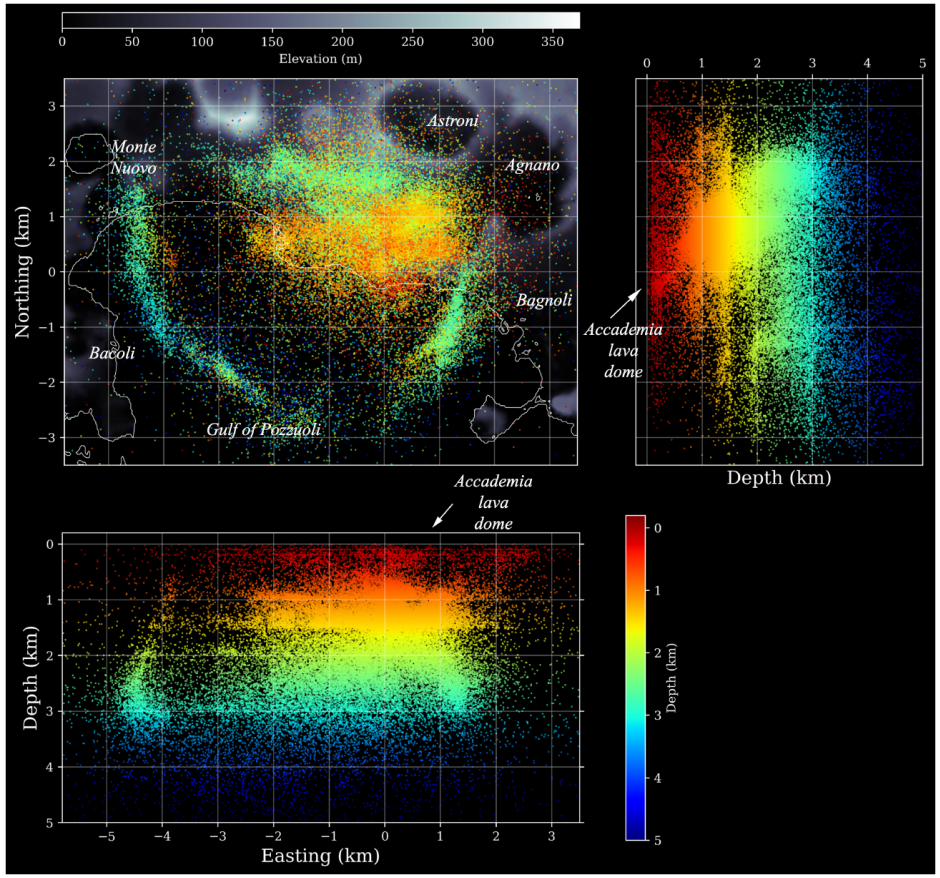
### Panel S5, all views — partially reproduced




External routine-catalog data, independently selected and plotted: 12,503 located events, 21 January 2022–20 March 2025. This is 420 more than the published historical snapshot. No author-derived catalog is used to adjust that difference; these are not independent detections.

# Figure S6: Initial VELEST locations

## Published reference




Tan et al. (2025), tanmodelsupp.pdf, p. 17.
DOI: 10.1126/science.adw9038.

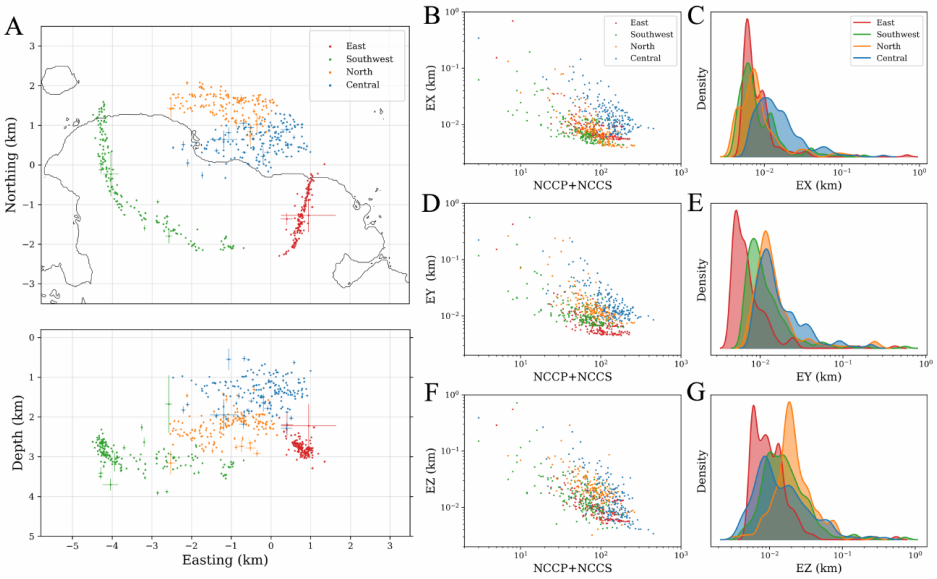
## Independent result

**All panels/views — not reproduced**

No accepted current result. For S6, the unverified background grid and dependent previous pilot were quarantined; alternative-profile arithmetic tests are not validated locations. No older output is substituted.

# Figure S7: Location uncertainties

## Published reference




Tan et al. (2025), tanmodelsupp.pdf, p. 18.
DOI: 10.1126/science.adw9038.

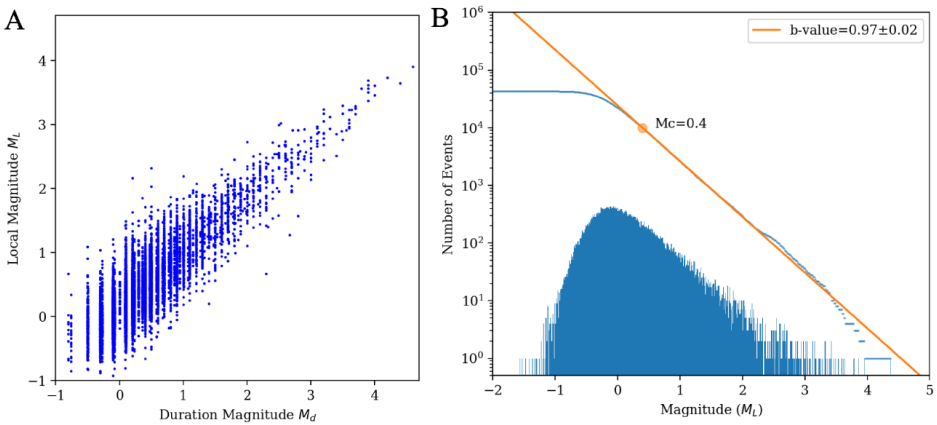
## Independent result

**All panels/views — not reproduced**

Panels A–G require independently selected regional subsets, HypoDD SVD locations and coordinate uncertainties. None of these scientific inversions ran.

# Figure S8: Magnitude calibration and frequency distribution

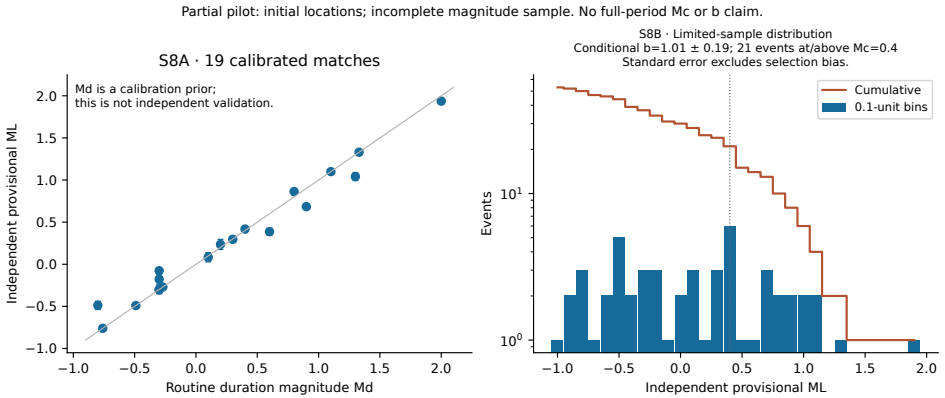
## Published reference




Tan et al. (2025), tanmodelsupp.pdf, p. 19.
DOI: 10.1126/science.adw9038.

## Independent result

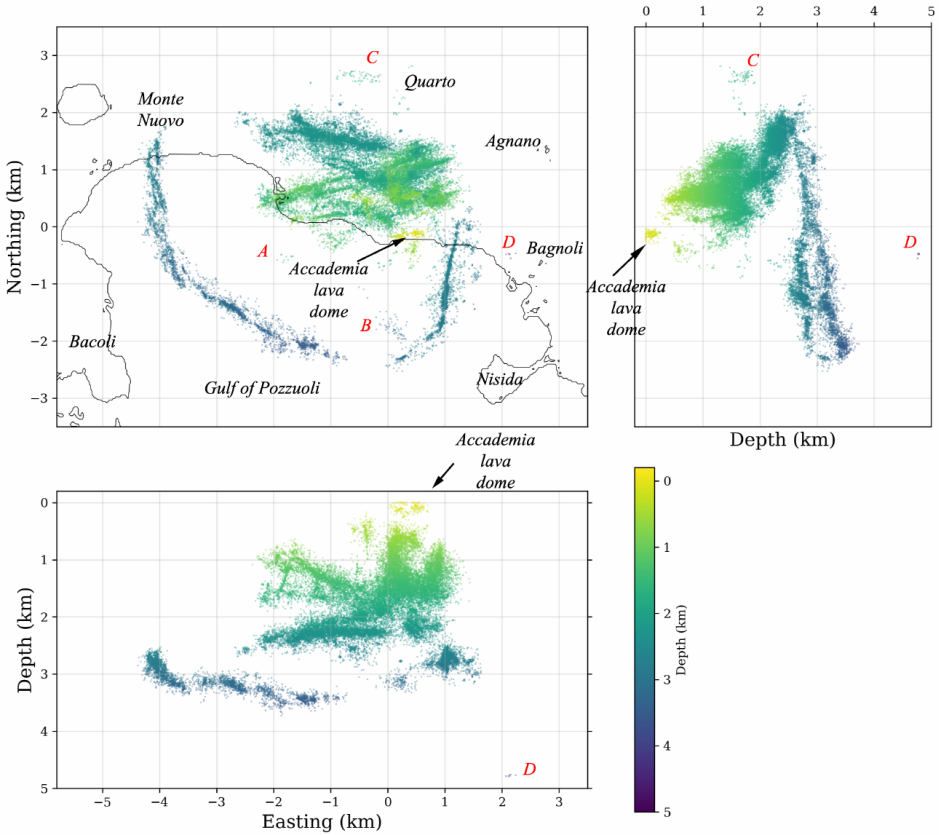
### Panel S8A–B — partially reproduced




Partial amplitude-ratio pilot: 53 provisional magnitudes from independent observations, with routine Md as the documented calibration prior. A is not an independent validation of that prior. B is a limited-sample histogram with a conditional likelihood estimate, not the full-period completeness or b-value result. Locations are initial associations, before final relocation.

# Figure S9: Best-located earthquake subset

## Published reference




Tan et al. (2025), tanmodelsupp.pdf, p. 20.
DOI: 10.1126/science.adw9038.

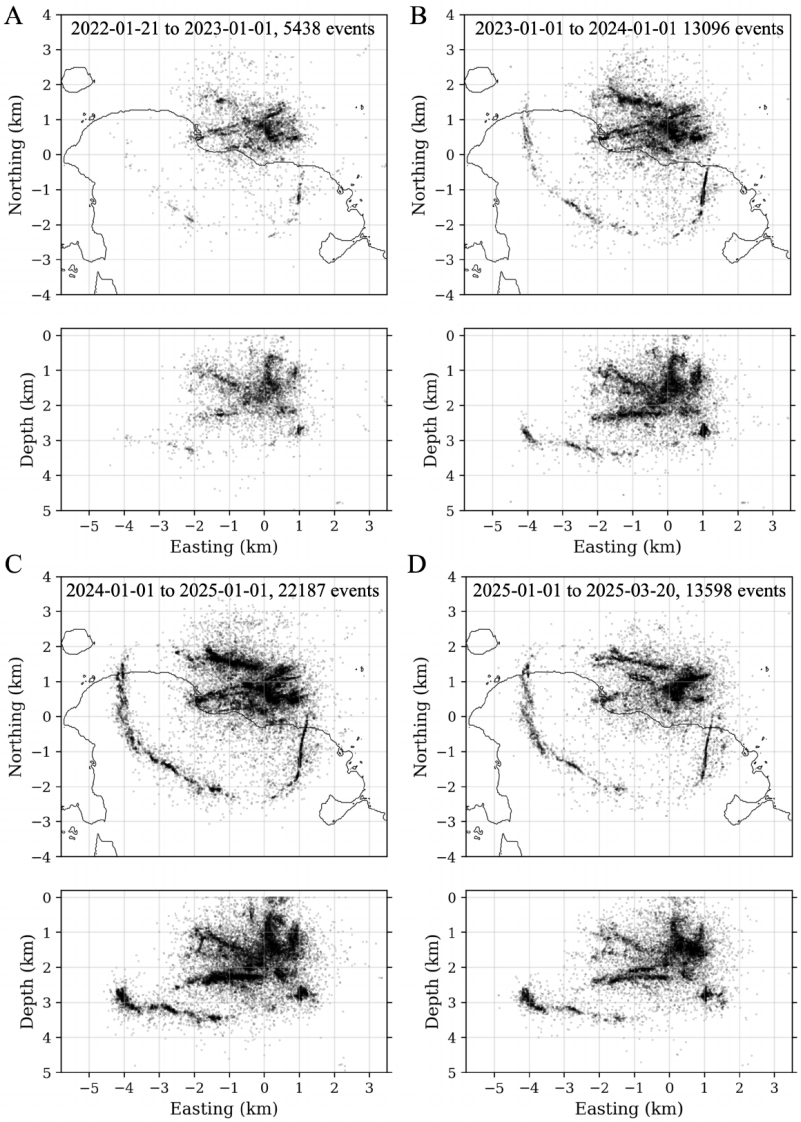
## Independent result

**All panels/views — not reproduced**

All three views require independently relocated events and cross-correlation observation counts for the published quality filter. These outputs are absent.

# Figure S10: Seismicity by year

## Published reference




Tan et al. (2025), tanmodelsupp.pdf, p. 21.
DOI: 10.1126/science.adw9038.

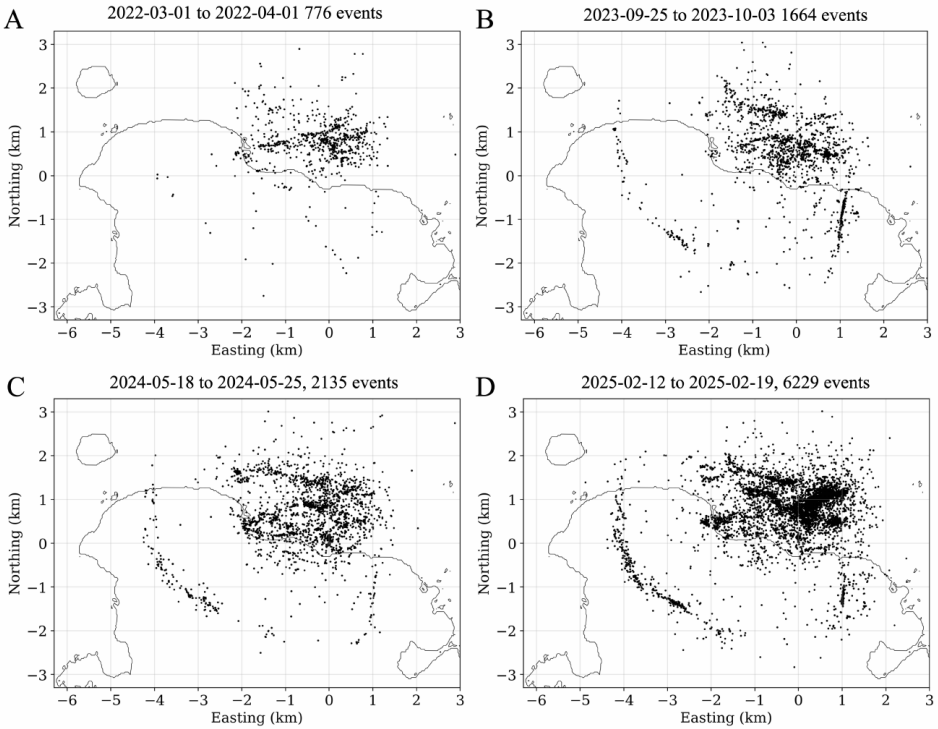
## Independent result

**All panels/views — not reproduced**

The four annual groups for 2022–2025 require the independent multi-year relocated catalog. The available sample does not support them.

# Figure S11: Four major swarms

## Published reference




Tan et al. (2025), tanmodelsupp.pdf, p. 22.
DOI: 10.1126/science.adw9038.

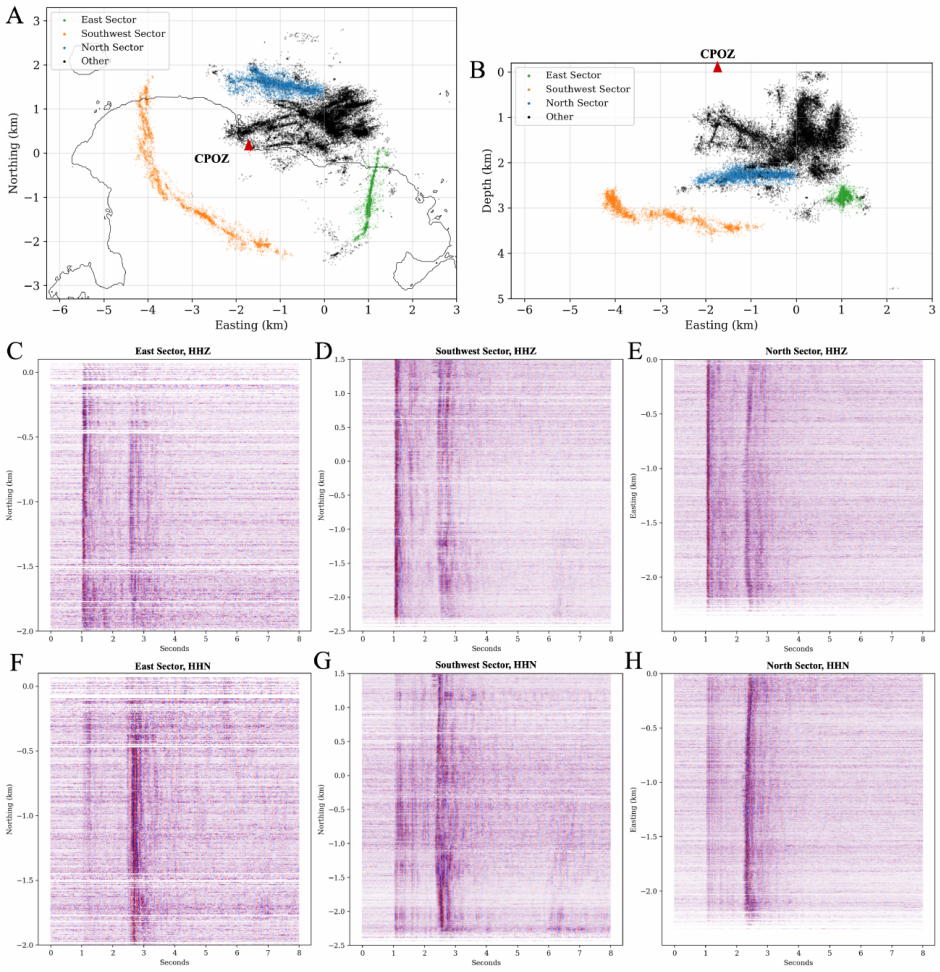
## Independent result

**All panels/views — not reproduced**

The four swarm maps require the independent relocated catalog and justified swarm windows. These selections and locations remain incomplete.

# Figure S12: Ring sectors and CPOZ common-receiver gathers

## Published reference




Tan et al. (2025), tanmodelsupp.pdf, p. 23.
DOI: 10.1126/science.adw9038.

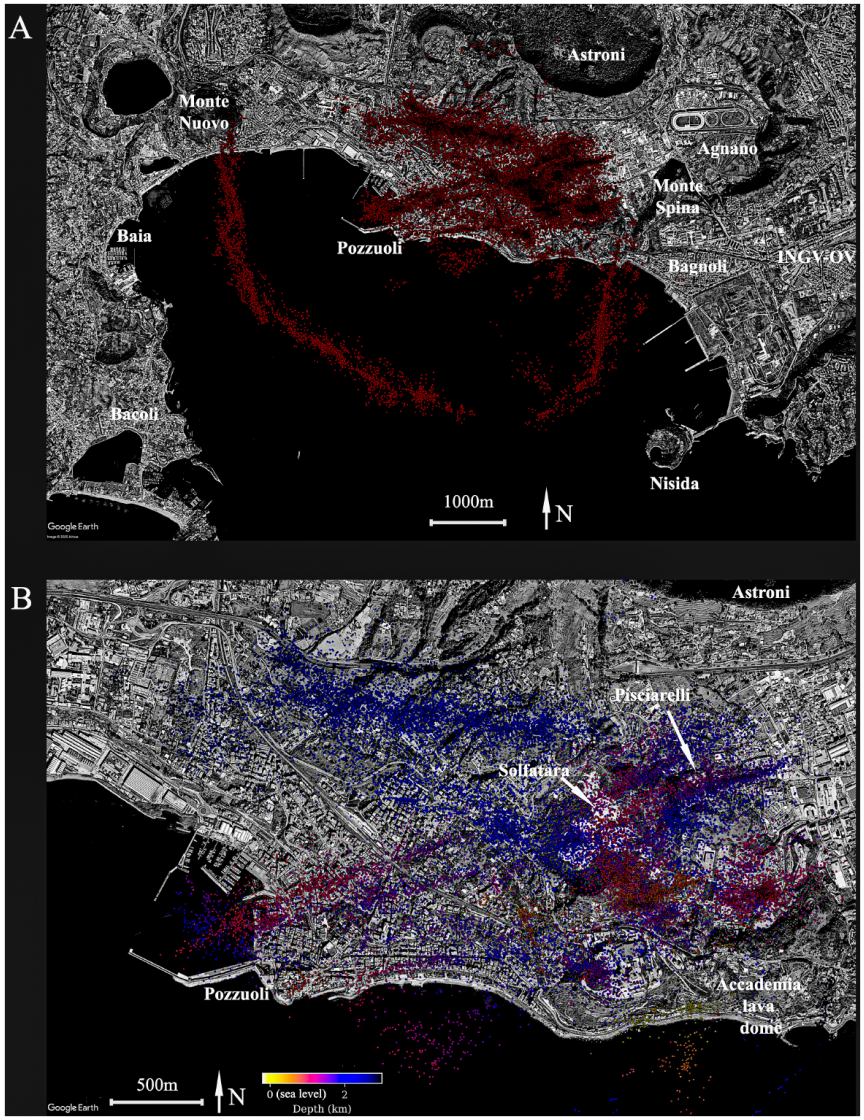
## Independent result

**All panels/views — not reproduced**

Panels A–H need relocated sector selections and manual depth review. 64 independent CPOZ P arrivals now exist, but the required sector gathers remain unfinished.

# Figure S13: Relocated seismicity on satellite imagery

## Published reference




Tan et al. (2025), tanmodelsupp.pdf, p. 24.
DOI: 10.1126/science.adw9038.

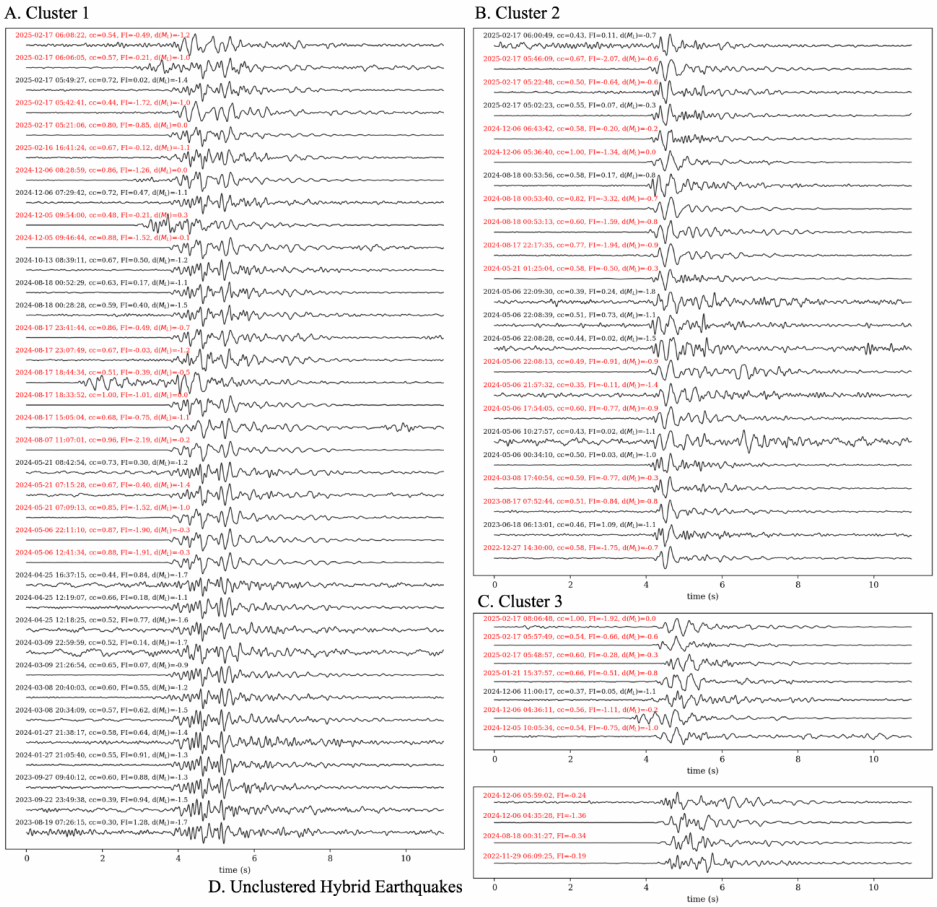
## Independent result

**All panels/views — not reproduced**

Panels A and B require reliable independent relocated coordinates and an appropriately sourced basemap. The relocated catalog does not exist.

# Figure S14: Hybrid-event clusters

## Published reference




## Independent result

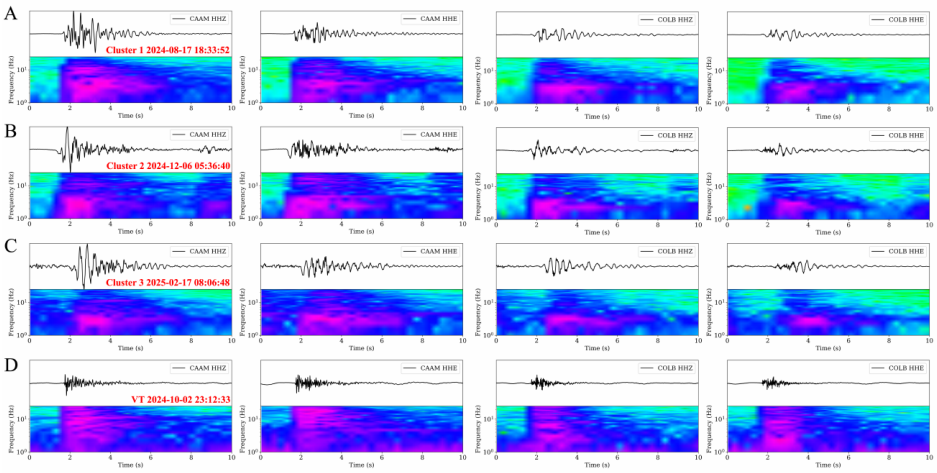
**All panels/views — not reproduced**

The cluster histories require three-year exposure, confirmed events and relative magnitudes. The present 75.797 hour scan produced 822 unconfirmed threshold crossings, not reviewed families.

Tan et al. (2025), tanmodelsupp.pdf, p. 25.
DOI: 10.1126/science.adw9038.

# Figure S15: Hybrid and shallow-event waveforms

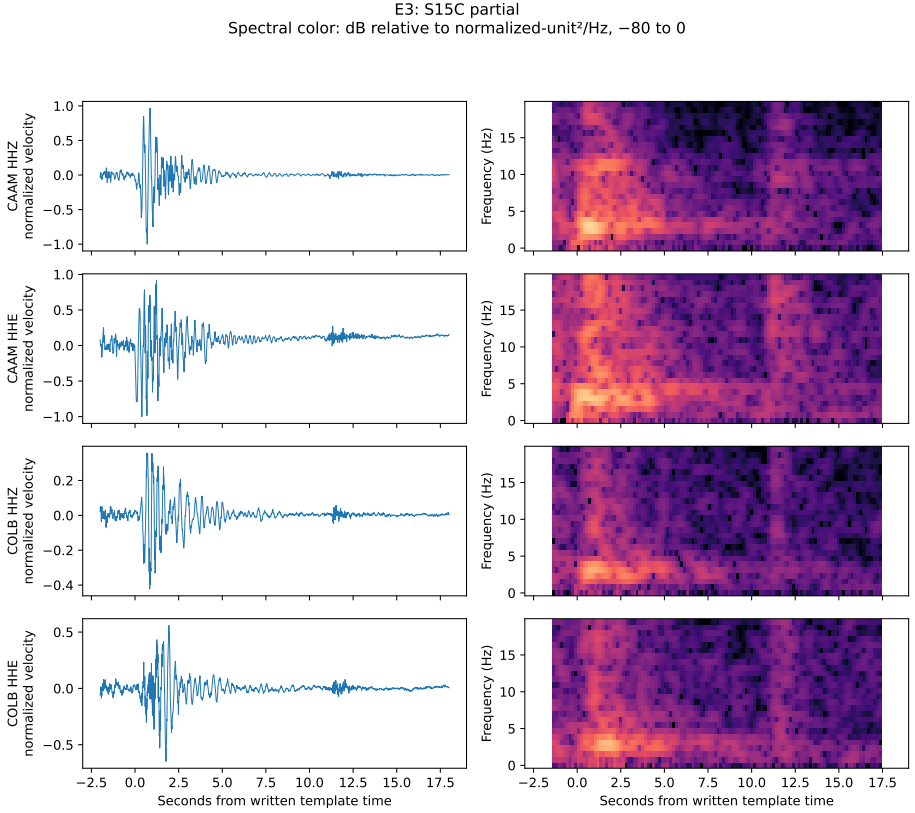
## Published reference




Tan et al. (2025), tanmodelsupp.pdf, p. 26.
DOI: 10.1126/science.adw9038.

## Independent result

### Panel S15C — partially reproduced




Panel C is partial: independently processed E3 CAAM/COLB observations. Onset alignment, display filtering and spectral settings differ or are incompletely specified. A, B and D have placeholders rather than substitute waveforms.

**Panels A, B and D — not reproduced**

Panels A and B use different published events from the saved E1/E2 templates; D has no independently selected matching example. Those panels remain not reproduced.